

[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

(Swap in your school's actual letterhead / logo here)

CA-1 — ANSWER KEY

Class : V

Max Marks : 30

Subject : Mathematics

Duration : 1 Hour

Syllabus: We the Travellers—I • Fractions • Angles as Turns

General Instructions: All questions are compulsory. Read each question carefully before answering.

Section A — 1 Mark Each

1. Which number has the same nearest ten and nearest hundred? (a) 7,126 (b) 7,835 (c) 6,999 (d) 7,030

Ans: (c) 6,999 (nearest ten = 7,000, nearest hundred = 7,000)

2. If $\frac{1}{3} = \frac{2}{x}$, find the value of x.

Ans: $x = 6$

3. Which fraction is the greatest: $\frac{5}{9}$, $\frac{5}{7}$, or $\frac{5}{12}$?

Ans: $\frac{5}{7}$

4. Starting from South, you make a full turn clockwise. Which direction do you end up facing?

Ans: South

Section B — 2 Marks Each

5. Choose the digits 1 and 9. Form two 2-digit numbers and find the result of the first subtraction only.

Ans: $91 - 19 = 72$

6. Explain in one line why the repeated-subtraction digit puzzle always ends at 9, no matter which two different digits are chosen.

Ans: Because the difference between any two-digit number and its digit-reversal is always a multiple of 9

7. For the digits 2 and 8, find (difference of digits) $\times 9$, and check it against $82 - 28$.

Ans: $(8-2) \times 9 = 54$; $82 - 28 = 54$, so they match

8. Compare $\frac{3}{9}$ and $\frac{4}{8}$ using two methods: (i) simplifying each fraction (ii) comparing with $\frac{1}{2}$.

Ans: $\frac{3}{9} = \frac{1}{3} < \frac{1}{2}$; $\frac{4}{8} = \frac{1}{2}$; so $\frac{4}{8} > \frac{3}{9}$ by both methods

9. Give one example of a number whose nearest hundred and nearest thousand are the same, and explain why this happens.

Ans: e.g. 7,030: nearest hundred = 7,000, nearest thousand = 7,000; this happens because the tens/ones part is small enough that both round down to the same value

10. If the minute hand turns $\frac{4}{12}$ of a full turn from 12, how many minutes does this represent, and what is $\frac{4}{12}$ in its simplest form?

Ans: $\frac{4}{12} \times 60 = 20$ minutes; simplest form of $\frac{4}{12}$ is $\frac{1}{3}$

Section C — 3 Marks Each

11. A cyclist travels at 15 km/hr and a motorbike at 50 km/hr. In 3 hours, how much farther does the motorbike travel than the cyclist? Show your working.

Ans: Cyclist: $15 \times 3 = 45$ km; Motorbike: $50 \times 3 = 150$ km; Difference = $150 - 45 = 105$ km

12. Rearrange the digits of 48,247 (using all 5 digits) to form the smallest possible 5-digit number and the largest possible 5-digit number. State both.

Ans: Smallest = 24,478; Largest = 87,442

Section D — 4 Marks Each (Word / Application Problems)

13. Town populations: Town 1 = 65,232; Town 3 = 56,380; Town 5 = 45,858; Town 6 = 66,540. (a) Arrange in descending order. (b) Round each to the nearest thousand. (c) State which towns' populations round up and which round down, with reasons.

Ans: (a) $66,540 > 65,232 > 56,380 > 45,858$. (b) 67,000; 65,000; 56,000; 46,000. (c) Town 1 (65,232) rounds down since hundreds digit $2 < 5$; Town 3 (56,380) rounds down since hundreds digit $3 < 5$; Town 5 (45,858) rounds up since hundreds digit $8 \geq 5$; Town 6 (66,540) rounds up since hundreds digit $5 \geq 5$

Section E — 2 Marks Each (From Your Textbook)

14. Take the digits 3 and 7. Form the two possible 2-digit numbers, subtract the smaller from the bigger, and repeat the process with the digits of each new result until you reach a 1-digit number. Show all your steps.

Ans: $73 - 37 = 36$; $63 - 36 = 27$; $72 - 27 = 45$; $54 - 45 = 9$. Final digit = 9

15. Rearrange the digits of 48,247 (using all 5 digits) to form the largest possible 5-digit number.

Ans: 87,442